

10 (short)

(a) Sketch the stress-strain responses for a linear elastic material and for a viscoelastic material under cyclic loading with zero mean stress. Give an example of a viscoelastic material. [3]

(b) The upper part of Fig. 8 shows three models (labelled A-C) for the stress-strain responses of viscoelastic materials. The elements of the models are elastic springs and viscous dashpots, with the respective stiffnesses E and viscosities η shown. The lower part of the figure shows the instantaneous application of a constant stress σ_0 at time $t = 0$, with three responses in the strain $\varepsilon(t)$ (labelled I-III). **By inspection**, match the models A-C to the appropriate strain response, I-III, explaining your reasoning in each case. Give expressions for the strains indicated (ε_1 , ε_2 , ε_3 , ε_4) in terms of the applied stress σ_0 and the properties of the model elements. (NB **Solution of the full response $\varepsilon(t)$ is not required**). [7]

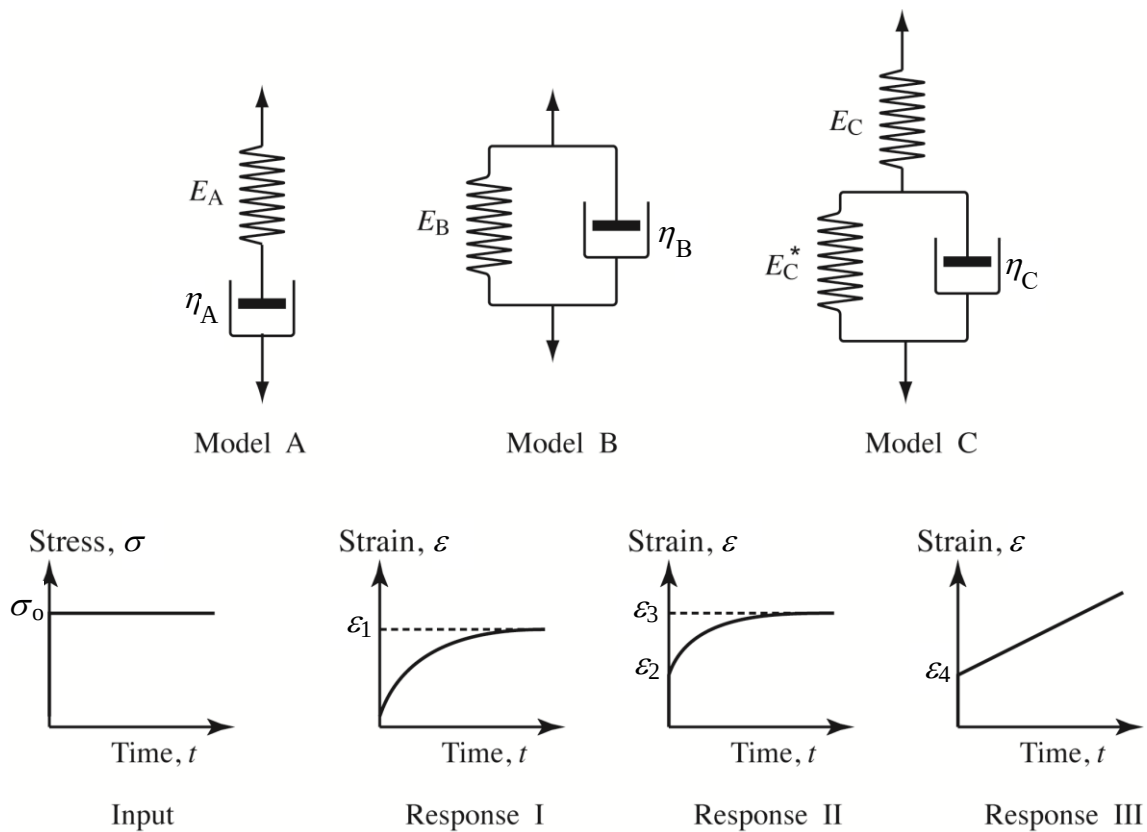


Fig. 8